

Lab 3 — Architecture Selection & Justification

Smart Lab Equipment & Slot Reservation Portal

Name: Rahul Devendra Kagawad

SRN: PES1UG24CS602

Sec: J

Architecture Selection

We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal.

Architectural Choice

The system is organized into Presentation, Business, and Data layers. The Presentation layer provides the user interface, the Business layer manages authentication, reservations and equipment operations, and the Data layer stores reservation and equipment information.

Reason 1 — Separation of Concerns

Layered architecture separates user interaction from reservation and equipment management logic. This makes the system easier to understand, maintain and modify. Changes to the user interface can be made without significantly affecting the reservation processing logic.

Reason 2 — Requirement Management

The Business layer can enforce important reservation rules such as the maximum 2-hour reservation duration, 7-day advance booking limit, equipment calibration validation, and prevention of overlapping reservations. The Data layer maintains reservation and equipment records. These rules correspond directly to the requirements of the Smart Lab system.

Security Advantage

The Authentication Service provides a security boundary before reservation operations. Only authenticated users can create or cancel reservations, preventing unauthenticated users from performing reservation operations. This satisfies the authentication requirement specified for the system.

Performance Benefit

The Reservation Manager can communicate directly with the Reservation Database for availability and overlap checks. Database-level locking can be used for concurrent reservation requests to prevent race conditions and double booking while meeting the required performance constraint.

Conclusion

Layered Architecture provides clear separation between presentation, business logic and data management, making it suitable for the Smart Lab Equipment & Slot Reservation Portal.